

Jorge Maldonado

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EDUCATION

John Jay College of Criminal Justice – CUNY

Bachelor of Science, Computer Science

New York, NY

May. 2027

Hostos Community College – CUNY

Associate of Science, Computer Science

New York, NY

May 2025

EXPERIENCE

Digital Literacy Instructor

Hostos Community College

May. 2025 – Aug. 2025

New York, NY

- Designed and delivered digital literacy curriculum to 30+ students covering software applications, internet safety, and digital communication tools
- Mentored 2 students through independent research projects from experiment design to data collection, each conducting surveys of 20+ participants and delivering final presentations

Scientific Research Student

CSTEP – Hostos Community College

Jan. 2023 – Sept. 2025

New York, NY

- Led design and execution of experiments comparing a recycled-material prototype refrigerator against a commercial equivalent, producing actionable sustainability insights
- Engineered an Arduino-based IoT system in C++ to autonomously record temperature and humidity data, eliminating manual data collection errors

IT Intern – Mexican Studies Program Scholarship

Bronx Community College – CUNY

Jan. 2024 – Mar. 2024

Bronx, NY

- Deployed and configured Windows workstations and implemented imaging/mirroring solutions to standardize lab configurations across multiple computer labs
- Provided tier-1 IT support by diagnosing and resolving hardware and software issues, minimizing downtime for end users

Peer Leader – C++ Programming

Hostos Community College

Jan. 2024 – Jun. 2024

New York, NY

- Coordinated and delivered C++ programming sessions to 30+ students, increasing hands-on project completion rates and coding proficiency
- Led group and one-on-one tutoring sessions reinforcing core OOP and data structure concepts

PROJECTS

CUNY Degree Helper | *Python, Streamlit, LangChain, ChromaDB, Anthropic Claude API* Aug. 2025 – Nov. 2025

- Built a RAG-powered academic planning tool using LangChain, ChromaDB, and sentence-transformer embeddings, deployed live to Streamlit Community Cloud with a full frontend/backend architecture
- Integrated Anthropic Claude API as an AI advisor to answer degree-related questions with context-aware responses from a persisted vector database
- Won 1st place at the CUNY GenAI Hackathon (2025), sponsored by Ernst & Young (EY); subsequently pitched project to investors for potential funding consideration

Asteroid Shooter Game | *Python, Pygame*

2024

- Engineered a 2D arcade-style asteroid shooter in Python using Pygame, structured across modular files with a custom `CircleShape` base class enabling reusable OOP collision detection via inheritance
- Developed a real-time scoreboard system and asteroid field spawner with randomized trajectories and split-on-hit mechanics

Mental Mirror | *Python, Streamlit, OpenAI API, Supabase, PostgreSQL*

Jul. 2025

- Built a full-stack AI-powered learning platform with 4 distinct features: AI journal analysis, Feynman technique practice, speech upload & transcription, and AI-generated feedback
- Implemented secure user authentication and per-user data isolation using Supabase with Row Level Security (RLS) and JWT token-based access control

- Architected a frontend/backend separation with OpenAI API integration for AI feedback, deployed live to Streamlit Community Cloud; garnered 10 GitHub stars

FitNature Hobby Hub | *React, JavaScript, Supabase, PostgreSQL, Vite, CSS*

2025

- Developed a full-stack React web application for organizing fitness and hobby workout groups, with a Supabase PostgreSQL backend for real-time data persistence
- Built responsive UI with CSS and Vite build tooling; implemented database schema with SQL migrations and Supabase client integration for group management features

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS, SQL, C++

Frameworks & Libraries: React, Streamlit, Pygame, LangChain, Pandas, Sentence Transformers, Vite

AI & Data: Anthropic Claude API, OpenAI API, ChromaDB, RAG, LLMs, Generative AI

Backend & Databases: Supabase, PostgreSQL, REST APIs, Row Level Security (RLS)

Developer Tools: Git, GitHub, VS Code, Arduino, Windows OS, Hardware Troubleshooting

CERTIFICATIONS

React Web Development – CodePath

Technical Interview Preparation – CodePath (LeetCode & Data Structures and Algorithms)